

CoreOps: Week 3 Module

Kubernetes Security & Platform Hardening

Objective: Master Kubernetes security fundamentals, pod-level & node-level isolation, RBAC correctness, secrets management, and supply-chain trust; the backbone of production-grade clusters.

TUESDAY: RBAC & Namespaces Basics

What You Should Understand Today

- What RBAC is
- What a Role / RoleBinding is
- Why namespaces matter in multi-team setups

Key Outcomes

- You can create a role with basic permissions
 - You can restrict a user/service account to a namespace
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WHAT TO PRACTICE

Lab 1: Create a Read-Only Role

Allow only “get, list, watch” on pods in a namespace.

Lab 2: Bind the Role

Attach the role to a user/service account.

Lab 3: Test Permissions

Run: `kubectl auth can-i create pods`

and observe.

WHERE TO STUDY

- RBAC basics doc
 - Namespace fundamentals
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INFRATHRONE NOTES

- RBAC misconfigurations are more common than container attacks.
- Start with the rule: “Give only what is required.”